

Fairness Without Demographics in Repeated Loss Minimization

Hashimoto et al., ICML 2018.

Presented by: Raunak Kumar

Cornell University

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Overview

- 1 Quiz
- 2 Introduction
 - Problem
 - Related Work
- 3 ERM and Disparity Amplification
- 4 Distributionally Robust Optimization
- 5 Experiments
- 6 Discussion

Question 1

Let

- $R(\theta)$ be the population risk,
- $R_k(\theta)$ be the risk for group k , and
- $R_{\max}(\theta)$ be the maximum of all group risks.

What is representation disparity?

- 1 High $R(\theta)$, low $R_{\max}(\theta)$
- 2 Low $R(\theta)$, high $R_{\max}(\theta)$
- 3 Low $R(\theta)$, high $\sum_k R_k(\theta)$
- 4 High $R(\theta)$, low $\sum_k R_k(\theta)$

Question 2

Recall the dynamics model presented in the paper. Which of the following is true?

- ① If we use ERM a fixed point is always stable.
- ② If we use ERM a fixed point is always unstable.
- ③ If we start at a fair fixed point and use ERM, we will converge to a different fair fixed point.
- ④ If we start at a fair fixed point and use ERM, we could converge to a different, potentially unfair, fixed point.

Motivation

- Speech recognizers are deployed to millions of users
- State of the art models achieve very high overall accuracy

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What could possibly go wrong?

- They have systematically high errors on minority accents
- Representation disparity: high overall accuracy, low minority accuracy
- Result of optimizing for optimizing average loss
- Same phenomenon observed in other applications

Goal

Consider the following setup

- Each user belongs to one of K groups
 - K , group memberships, and group distributions are unknown
- At each time step
 - Each user interacts with ML system and incurs some loss
 - Based on loss they decide to keep using the system or quit
 - Model is retrained on user data and used in the next time step

Goal: Control the worst case risk of any group across time

Overview

- Define fairness criteria and dynamics model
- Empirical risk minimization (ERM) does not control the worst case risk over groups
- Distributionally robust optimization (DRO) remedies this issue
 - Let $\alpha_{\min}$ be a lower bound on the group proportion
 - Optimize worst case risk over a χ^2 divergence ball
 - This bounds the worst case risk over groups
 - Computationally efficient and widely applicable
- Experiments
 - Toy examples
 - Real world Amazon Mechanical Turk task

Related Work

- John Rawls' theory on distributive justice and stability
 - View predictive accuracy as a resource to be allocated
- Fairness for classification involving protected labels
 - Constraints - equalized odds, calibration, etc.
 - Requires use of demographic labels
 - Full demographic data is often unavailable in speech recognition
- In an online setting static fairness constraints do not ensure fairness over time

Representation Disparity

- User makes a query $Z \sim P$, model $\theta \in \Theta$ makes a prediction
- User incurs loss $\ell(\theta; Z)$
- Risk or expected loss $R(\theta) = E_{Z \sim P}[\ell(\theta; Z)]$
- Assume $Z \sim P := \sum_{k \in [K]} \alpha_k P_k$
 - K , α_k and P_k are all unknown
- Risk for group k $R_k(\theta) = E_{P_k}[\ell(\theta; Z)]$

Goal: Control $R_{\max}(\theta) = \max_{k \in [K]} R_k(\theta)$

Representation disparity: low $R(\theta)$, high $R_{\max}(\theta)$

Retention Rate

Definition

Retention rate (or function) ν is a differentiable, strictly decreasing function that maps a risk level R to the fraction of users who continue to use the system.

For example,

- Let R_k be the expected zero-one loss
- Let $\nu(x) = 1 - x$
- Then $\nu(R_k(\theta)) \rightarrow$ users leave proportional to the misclassification rates of their queries.

Dynamics

Definition

Given a sequence θ^t , for each $t \in [T]$, the expected number of users λ^t and samples Z_i^t starting at $\lambda_k^0 = b_k$ is governed by:

$$\text{(Group size)} \quad \lambda_k^{t+1} = \lambda_k^t \nu(R_k(\theta^t)) + b_k$$

$$\text{(Group proportion)} \quad \alpha_k^{t+1} = \frac{\lambda_k^{t+1}}{\sum_{k' \in [K]} \lambda_{k'}^{t+1}}$$

$$\text{(Number of users)} \quad n^{t+1} = \text{Poisson} \left(\sum_{k \in [K]} \lambda_k^{t+1} \right)$$

$$\text{(Samples)} \quad Z_1^{t+1}, Z_2^{t+1}, \dots, Z_{n^{t+1}}^{t+1} \sim P^{t+1} = \sum_{k \in [K]} \alpha_k^{t+1} P_k$$

- ERM defines parameter sequence as $\theta^t = \arg \min_{\theta \in \Theta} \sum_i \ell(\theta; Z_i^t)$
- Goal is to control $R_{\max}^T(\theta^0, \dots, \theta^T) = \max_{k, T} R_k(\theta^t)$

Challenges

- Without demographics we can't even measure $R_{\max}^T$, let alone minimize it
- Need to ensure group proportions are stable
 - If $\alpha_k^t \rightarrow 0$ as $t \rightarrow \infty$ then the group has near zero probability of appearing in samples

Fixed Points and Fairness

- View the expected user counts λ^t as a dynamical system
- Long-term fairness criteria controlled by
 - Does λ have a fair fixed point?
 - Fair = population fraction where risk minimization maintains the same population fraction over time
 - Is this fixed point stable?

Models that are initially fair with low representation disparity may become unfair over time if we use ERM!

Disparity Amplification with ERM

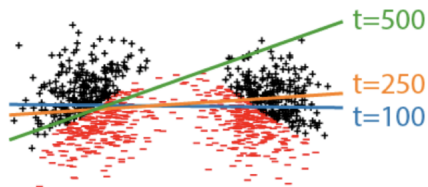


Figure 1. An example online classification problem which begins fair, but becomes unfair over time.

Conditions for Disparity Amplification

Definition

Let Φ be the update for the expected population size

$$\begin{aligned}\lambda_k^{t+1} &= \Phi(\lambda_k^t) = \lambda_k^t \nu(R_k(\theta(\lambda_k^t))) + b_k \\ \theta(\lambda_k^t) &= \arg \min_{\theta} E_{\sum_k \alpha_k^t P_k}[\ell(\theta; Z)]\end{aligned}$$

- Arrival intensity λ^* is a fixed point if $\lambda^* = \Phi(\lambda^*)$.
- Fixed point is stable when the maximum modulus of the eigenvalues of the Jacobian of Φ is less than one and unstable whenever it is greater than one.
- Major quantities which control risk are
 - Retention rate ν and its derivative
 - $K \times K$ matrix $\nabla L H_R(\alpha^*)^{-1} \nabla L^T \approx$ changes in one group's risk as a function of another

Conditions for Disparity Amplification

Proposition 1. Let $\lambda^* = \Phi(\lambda^*)$ be a fixed point, and $\theta^* = \arg \min_{\theta} \mathbb{E}_{\sum_k \alpha_k^* P_k} [\ell(\theta; Z)]$ be the minimizer at λ^* .

Define $H_{\mathcal{R}}(\alpha^*)$ as the positive definite Hessian of the expected risk at θ^* , λ^* and define ∇L as the per-group parameter gradients at θ^* ,

$$\nabla L = \begin{bmatrix} \nabla_{\theta} \mathbb{E}_{P_1} [\ell(\theta^*; Z)] \\ \vdots \\ \nabla_{\theta} \mathbb{E}_{P_k} [\ell(\theta^*; Z)] \end{bmatrix}.$$

The arrival intensity λ^* is unstable whenever

$$\rho_{\max} \left(\text{diag}(\nu(\mathcal{R}(\theta(\lambda^*)))) - \text{diag}(\lambda^* \nu'(\mathcal{R}(\theta(\lambda^*))) \right. \\ \left. \nabla L H_{\mathcal{R}}(\alpha^*)^{-1} \nabla L^{\top} \left(\frac{I}{\sum_k \lambda_k^*} - \frac{\mathbf{1} \lambda^{*\top}}{(\sum_k \lambda_k^*)^2} \right) \right) > 1.$$

Conditions for Disparity Amplification

- Suppose we start at a fair fixed point $\lambda_1^* = \dots = \lambda_k^*$ and $R_1 = \dots = R_k$
- If decreasing the risk for one group increases the risk for others sufficiently
 - The fixed points is unstable
 - Model eventually converges to a different, possibly unfair, fixed point

Conditions for Disparity Amplification

Corollary 1 (Counterexample under symmetry). *Let $\lambda_1^* = \dots = \lambda_k^*$ be a fixed point with $\mathcal{R}_1 = \dots = \mathcal{R}_k$, then for any strongly convex loss,*

$$\rho_{\max} \left(\nabla L H_{\mathcal{R}}(\alpha^*)^{-1} \nabla L^\top \right) > \frac{1 - \nu(\mathcal{R}_1)}{-\nu'(\mathcal{R}_1)/k}. \quad (3)$$

is a sufficient condition for instability.

- LHS is maximum excess risk that can be incurred due to a small perturbation in the mixture weights
- RHS is the underlying stability of the dynamics and measures the sensitivity of λ with respect to risk

Mean and Median Estimation

Consider the following mean estimation example

- Each user belongs to one of 2 groups, -1 or 1
- Incurs loss $(\theta - Z)^2$
- $\theta = 0$ is a fair fixed point since it equalized losses to both groups
- $H_{risk}(\alpha^*) = 1/2$ and $\nabla L = [2, -2]$. Thus, LHS is 4
- If $\nu(x) = \exp(-x)$ then RHS is approximately 3.4
- Any perturbation will eventually lead to $\lambda_1 \neq \lambda_2$
- All other fixed points are unfair solutions (mean of one of the groups)

Mean and Median Estimation

Consider the following median estimation example

- Non-unique minimizer at 0 when $\lambda_1 = \lambda_2$
- If $\lambda_1 > \lambda_2$ then minimizer becomes -1
- No retention function ν can fix this

Bounding the Risk

- Denote the χ^2 divergence between probability distribution Q and P as

$$D_{\chi^2}(Q\|P) = \int \left(\frac{dQ}{dP} - 1 \right)^2 dQ$$

- Let $B(P, r)$ be the chi-squared ball around P of radius r

$$B(P, r) = \{Q \ll P : D_{\chi^2}(Q\|P) \leq r\}$$

- Consider worst case loss over this ball

$$R_{dro}(\theta; r) = \sup_{Q \in B(P, r)} E_Q[\ell(\theta; Z)]$$

Bounding the Risk

Proposition

For $P = \sum_{k \in [K]} \alpha_k P_k$, we have $R_k(\theta) \leq R_{dro}(\theta; r_k)$ for all $\theta \in \Theta$ where $r_k = (1/\alpha_k - 1)^2$ is the robustness radius.

- For each group k $R_k(\theta) \leq R_{dro}(\theta; r_k)$.
 - Choice of r_k ensures P_k is inside the chi-squared ball around P .
- Thus, $R_{\max}^T \leq R_{dro}(\theta; r_{\max})$ where $r_{\max} = (1/\alpha_{\min} - 1)^2$.
- Bound is tighter when variation in loss is lower within groups than across groups.

Bounding the Risk

Proof.

Since P_k is a mixture component of $P = \sum_{i \in [K]} \alpha_i P_i$, we have that

$\frac{P(x)}{P_k(x)} = \alpha_k + \sum_{i \in [K] \setminus k} \alpha_i \frac{P_i(x)}{P_k(x)} \geq \alpha_k$. Consider the χ^2 -divergence between P_k and P

$$\begin{aligned} D_{\chi^2}(P_k \| P) &= \int_x \left(\frac{P_k(x)}{P(x)} - 1 \right)^2 P(x) dx \\ &\leq \int_x \left(\frac{1}{\alpha_k} - 1 \right)^2 P(x) dx \\ &= r_k. \end{aligned}$$

This shows that for the given choice of r_k P_k is in the chi-squared ball around P . Since the DRO risk, $R_{\text{dro}}(\theta; r_k)$, is over all distributions Q in the ball, the upper bound holds. □

Dual

Intuitively, the DRO risk upweights examples with high loss as illustrated by the dual of the maximization problem:

Proposition

If $\ell(\theta; \cdot)$ is upper semi-continuous for any θ , then for $r_{\max} \geq 0$ and any θ , $R_{\text{dro}}(\theta; r_{\max})$ is equal to the following

$$\inf_{\eta \in \mathbb{R}} \{F(\theta; \eta) = C(E_P[[\ell(\theta, Z) - \eta]_+^2])^{1/2} + \eta\}$$

where $C = (2(1/\alpha_{\min} - 1) + 1)^{1/2}$.

Median Estimation

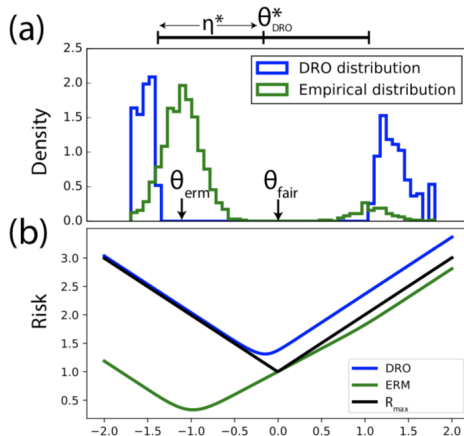


Figure 2. Chi-square distributionally robust optimization (DRO) regularizes the losses (top panel) such that the minimum loss estimate is fair to both groups (bottom panel).

Optimization

How do we minimize $R_{dro}(\theta; r_{\max})$ over θ efficiently?

Option 1:

- For a given η compute approximate minimizer $\hat{\theta}_n$ of $E_P[\ell(\theta; Z) - \eta]_+^2$
- We have $R_{\max}(\hat{\theta}_n) \leq R_{dro}(\hat{\theta}_n; r_{\max}) \leq F(\hat{\theta}_n; \eta)$
- We can treat η as a hyperparameter
- For convex losses l , F is also convex $\rightarrow$ we can binary search over η

Optimization

How do we minimize $R_{dro}(\theta; r_{\max})$ over θ efficiently?

Option 2:

- Suppose we can compute $\theta^*(Q) \in \arg \min_{\theta \in \Theta} E_Q[\ell(\theta; Z)]$ efficiently
- Compute worst case $Q^*(\theta) \in \arg \max_{Q \in B(P, r)} E_Q[\ell(\theta; Z)]$ for a given θ using projected gradient descent
- Alternate optimization on θ and Q to find saddle point (θ^*, Q^*) such that $\theta^* = \theta^*(Q^*)$ and $Q^* = Q^*(\theta^*)$

Stability

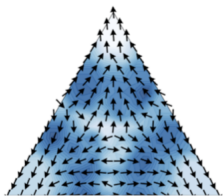
Proposition 4. Assume the retention model in Definition 1. Let $\alpha_k^{(t)} > \alpha_{\min}$, $\frac{b_k}{\sum_k b_k} > \alpha_{\min}$, $\lambda^{(t)} := \sum_k \lambda_k^{(t)} \leq \frac{\sum_k b_k}{1 - \nu_{\max}}$, and $\nu(\mathcal{R}_k(\theta^{(t)})) < \nu_{\max}$. Then, whenever we have

$$\mathcal{R}_k(\theta^{(t)}) \leq \nu^{-1} \left(1 - \frac{(1 - \nu_{\max})b_k}{\alpha_{\min} \sum_k b_k} \right),$$

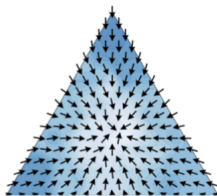
$$\alpha_k^{(t+1)} = \frac{\lambda^{(t)} \alpha_k^{(t)} \nu(\mathcal{R}_k(\theta^{(t)})) + b_k}{\sum_l \lambda^{(t)} \alpha_l^{(t)} \nu(\mathcal{R}_l(\theta^{(t)})) + b_l} > \alpha_{\min}.$$

Thus, if the risk at each time step is small enough then optimizing the DRO risk at each time step controls the unknown worst case group risk over all time steps.

Stability



(a) ERM



(b) DRO ($\alpha_k = 0.4$)

Figure 3. Dynamics of repeated median estimation - shading indicates velocity at each point. ERM results in unfair parameter estimates that favor one group. DRO is strongly stable, with an equal proportion groups being the unique stable equilibrium.

Simulated Task

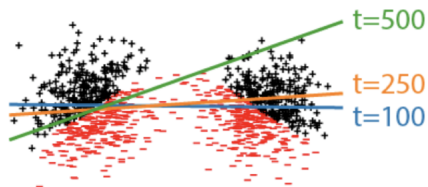


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Simulated Task

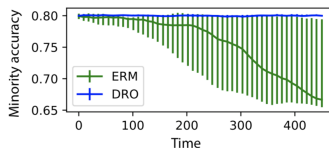


Figure 5. Disparity amplification in Figure 1 is corrected by DRO. Error bars indicate quartiles over 10 replicates.

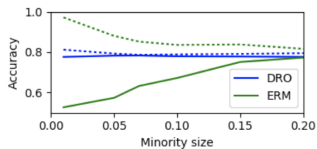


Figure 6. Classifier accuracy as a function of group imbalance. Dotted lines show accuracy on majority group.

Autocomplete Task

- Task: Prediction of next words
- Corpus of tweets from two estimated demographic groups: African Americans and White Americans
 - AAE: African American English, SAE: Standard-American English
 - Several distinguishing linguistic features between the two

Autocomplete Task

- Train 5 models with different fractions of tweets labelled as AAE
- A user can be assigned to 10 different systems.
 - Simulates user from a particular demographic group
- Evaluate retention and loss for AAE and SAE separately
 - User is assigned 10 tweets from a held out test set of AAE or SAE tweets
 - Need to replicate using a web-based keyboard along with autocomplete system
- Fill out survey at the end of the task
- Total of 1000 users' feedback: 50 users for each of two held out test set types and one of 10 models

Autocomplete Task

Instructions

We would like you to evaluate a text auto-completion system and your likelihood of using such a system. We will ask you to use a virtual keyboard with an auto-completion system to re-type 10 Tweets from Twitter, using the auto-complete to help you as you wish. You will have 1 hour to complete this task. *Note that this task works best with those with fast network speed. You can check whether your current network speed is fast enough at this link:*

<http://34.215.89.154:8080/static/latency.html> (<http://34.215.89.154:8080/static/latency.html>)

DISCLAIMER: As we will be asking you to re-type real data from Twitter, the example texts in this system may contain offensive text and slang, despite automated filtering.

After following the instructions below, please answer the questions in the survey:

1. Click on the "START TASK" link after the instructions.
2. Next to the "Please type this" entry is the text you have to replicate using the keyboard below. Start by clicking the first letter of the text on the keyboard.
3. If there exists any auto-complete suggestions, they appear as a drop down menu. Use them in the way that is most helpful for you.
4. If you enter a wrong key, the mismatch will be displayed next to the "Difference" entry. Use this entry to guide you to complete replicating the text.
5. Once you have successfully replicated, the accept button will turn green. Click to receive the next tweet.
6. Repeat until you have successfully replicated 10 tweets. Then, cancel the window and fill out the survey questions below.

We will manually inspect every answer before accepting your work, and will be logging your entries on the web form to verify you correctly replicated 10 tweets. **Incomplete survey answers will not be approved.**

In addition, completion of the task will result in a unique code we ask you to enter at the end of this survey - no task will be approved without a correctly provided code. This task is expected to take 10 minutes to complete, but you will be allotted an hour.

INTERFACE LINK: If the link above is not shown there is an error with Javascript on your computer. To perform this HIT, you must have Javascript and cookies enabled on your browser.

Autocomplete Task

User: anonymous Remaining: 10

Please type this: my life has really changed I have become a true boss I

Difference: my life has really changed I have become a true boss I

my life haha

my life hahaha

my life has

my life have

my life here

1 2 3 4 5 6 7 8 9 0 Bksp

Tab Q W E R T Y U I O P

A S D F G H J K L Accept

Shift Z X C V B N M Shift

Clear

Figure 8. Autocomplete task interface on Amazon mechanical turk.

Autocomplete Task

In the following survey, please provide feedback on the suggestions and usefulness of the autocomplete system itself – NOT the interface. Any feedback regarding the interface can be states at the end.

1. Please describe good aspects of the autocomplete word suggestions:

2. On a scale of 1 (very unsatisfied) to 5 (extremely satisfied), how satisfied were you with the autocomplete suggestions?

- ☐ 1 - very unsatisfied, did not find autocomplete system helpful
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5 extremely satisfied, found autocomplete system helpful with task

3. What were some drawbacks you experienced with the autocomplete word suggestions?

4. On a scale of 1 (not frustrated) to 5 (extremely frustrated), how frustrated were you with the autocomplete suggestions?

- ☐ 1 - not frustrated at all
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5 - extremely frustrated

5. Would you use the suggestions from this autocomplete system to help with your own writing task?

- ☐ Yes
- ☐ No

Figure 9. Survey, page 1, Q1-Q4 are quality control verification questions. Q5 measures retention

Autocomplete Task

6. On a scale of 1 (very unhappy) to 5 (very happy), how happy were you with this hit?

- ☐ 1 - very unhappy
☐ 2
☐ 3
☐ 4
☐ 5 - very happy

7. Please provide feedback on the interface for this task:

8. Please provide your unique code given at the end of the task:

Figure 10. Survey, page 2. Q2 measures satisfaction, Q7 is used to measure issues with the HIT, and Q8 is used to ensure users completed

Autocomplete Task

- User satisfaction provides samples $\nu(R_k(\alpha))$ and surrogate for $R_k(\alpha)$.
- Interpolate ν and R to $\alpha \in [0, 1]$ to simulate dynamics
- Use bootstrap to estimate variability in estimates

Autocomplete Task

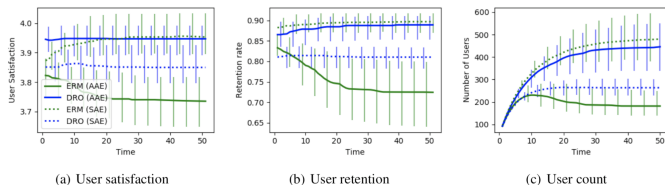


Figure 4. Inferred dynamics from a Mechanical Turk based evaluation of autocomplete systems. DRO increases minority (a) user satisfaction and (b) retention, leading to a corresponding increase in (c) user count. Error bars indicates bootstrap quartiles.

Note: slight mismatch between training loss and human satisfaction with autocomplete system.

Recap

- Criteria for fairness
 - Representation disparity
 - Disparity amplification
- ERM often results in unfairness
- DRO
 - Upper bounds the risk for minority groups
 - Easy to implement
 - Induces distributional robustness

Open Questions

- DRO can only control limited aspects of fairness
 - DRO implies certain minority groups
 - Some applications, such as loans, have well-specified demographic groups
 - These two may be very different
- Adversarial or high-noise settings
 - High loss, adversarial group can resemble a minority
 - Loosens DRO upper bound.
 - Algorithms that are fair to unknown latent groups and fair?

References



Tatsunori B. Hashimoto, Megha Srivastava, Hongseok Namkoong, Percy Liang
Fairness Without Demographics in Repeated Loss Minimization
ICML, 2018.

The End